

Project Initialization and Planning Phase

Date	25 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement):

Floods are one of the most destructive natural disasters, causing significant loss of life, damage to infrastructure, disruption of agriculture, and economic loss. Traditional flood forecasting methods are often slow or unable to effectively analyze multiple environmental factors. This project proposes a machine learning-based flood prediction system that uses historical weather and rainfall data to predict the likelihood of flooding and provide early warnings for improved disaster preparedness.

I AM	I'M TRYING TO	BUT	BECAUSE	WHICH MAKES ME FEEL
Regional Safety Officer or Meteorologist	Predict flood risks and log warning alerts in real-time	SQLite file-locking contentions crash the app under heavy user load	Default database configurations block concurrent write transactions	Anxious about missing critical early warnings during monsoons

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A disaster response coordinator.	Deploy rescue teams and medical resources ahead of time.	I do not have accurate, localized predictions of where floods will hit hardest.	Current weather forecasts only provide general rainfall data, not specific flood risks based on historical patterns.	Anxious and always one step behind the disaster.

PS-2	A local resident in a flood-prone zone.	Protect my family, property, and assets before the monsoon season peaks.	I receive warnings too late, or they are too vague to take meaningful action.	There is no accessible, data-driven web app alerting the community with enough lead time.	Helpless and frustrated by the unpredictability.
PS-3	An urban planning official.	Improve city infrastructure to prevent severe waterlogging.	I lack clear insights into which specific neighborhoods are statistically most vulnerable.	We do not have a predictive model analyzing years of historical weather and flood occurrences.	Inefficient and worried about future infrastructural damages.
PS-4	A regional meteorologist or safety officer.	Monitor regional meteorological metrics and predict high-risk flood occurrences.	Raw indicators are on different scales and SQLite database locks under high request load.	Default SQLite rollback configurations block write transactions when concurrent users request history alerts.	Anxious about delayed emergency responses and alert latency during monsoon periods.